

MAE C163C / C263C Project Report Guide

(Due online by 11:59pm PT on Saturday, 6/13)

- **Only one project report needs to be submitted per project team.** Each team member must write about some aspect of the project, preferably the portion that they worked on so that I can gauge effort, understanding, and clarity of presentation on an individual basis.
 - Remember, this course is not about mechanical design or trajectory planning. This course is about controller design. Be sure to highlight and demonstrate your understanding of feedback controls concepts for robotic systems.
- **The report should be formatted in the style of a technical IEEE conference paper with an abstract and keywords** (<https://www.ieee.org/conferences/publishing/templates.html>).
 - I expect the report to be written with the same attention to detail one would give a technical conference paper submission. That is, use technical terms and not jargon, use correct grammar, use a spell-checker, cite sources appropriately, create clearly labeled figures, make sure your figures/tables/equations are referenced from within the body of the text, etc. Do not assume that I will be familiar with your robot or concepts that were not covered in this course.
- **The report should contain the following items:**
 1. **Project title**, team #, team member names in alphabetical order by last name
 2. **Overview** of project, including a literature review (Why should anyone be interested?; Where might such a robot be used and for what purpose?)
 3. **Objectives of study and/or robot controller design**
 4. **Methods**
 - A visualization of your robot, either in simulation or via photograph.
 - If you used open source URDF files for your robot, you must cite their source.
 - Define and justify your controller design requirements. These requirements should be used to specify quantifiable performance metrics for your controller.
 - State all assumptions explicitly.
 - Highlight the application of concepts learned in class.
 - Include numbered equations and make sure they are referenced from the body of the text.
 5. **Results**
 - A video of your controller(s) in action is REQUIRED.
 - If you simulated your robot, include relevant snapshots of an animation.
 - If you implemented your controller on robot hardware, include relevant snapshots of a video of the robot.
 - An abbreviated series of time-lapse images can be used to show the robot's controlled motion, as applicable.
 - An error / performance assessment is REQUIRED.
 - Show clear, labeled time history plots of relevant controlled variables and performance metrics. Use colors, linetypes, and legends to make clear figures. Axis labels should include units. Do not assume that I will understand your jargon and symbols.
 - Do not include figures in your report simply because your code spits them out. You must have a clear purpose for including a figure in your report and you must comment on each figure. If you have a bunch of figures that all make the same point, please do not subject the reader to all of them. Pick the most representative figures for inclusion and discussion.
 - Include horizontal (dashed) lines for zero error so it is easy to see how your values compare.
 - If relevant, report joint angles and joint angle error plots in degrees, which are more intuitive than radians.
 - Be sure to compare computer simulations to actual experimental data whenever possible.

6. Discussion

- You must address the effects of your controller gain choices. What insights were learned during gain tuning?
- If you implemented multiple controllers of different types or if you tested different controller gains using the same controller architecture, you must compare and contrast the controller designs and/or gain combinations and their resulting performance. Which controller would you recommend and why?
- Be sure to address the capabilities and limitations of your controller(s). Take your controllers to the limit and see how they break.
- Describe challenges you faced during the project. Challenges are what make these controls problems interesting.

7. Future directions

8. Bibliography

- Be sure to cite the source of images and literature references, preferably from peer-reviewed technical journals. Citations need to appear within the text, when appropriate, and not just as a list at the end of the report.
- Treat the presentation and report as if you were presenting these materials for acceptance at a technical conference. *Wikipedia references for technical content are UNACCEPTABLE.*

9. Appendix

- This section does *not* need to be formatted in the style of IEEE conference proceedings.
- [An itemized breakdown of team members and specific contributions to the project is REQUIRED.](#) Each team member should be accountable for an equal amount of work. All team members should have been involved in the controller design process.
- Bill of materials with quantities, model numbers, vendors, and costs, if applicable. Include a total cost, if applicable.
- Custom CAD designs
- Custom code (e.g. Python, MATLAB, Simulink models)
- Custom systems-level wiring diagrams (e.g. for Arduino, DC servo motors, etc.)
- **Captioned figures and tables should be included and referenced from within the body of the text when appropriate.** Do not just paste in any and all figures/tables to pad your document! I will not take the time to sift through endless figures to try to extract their meaning. As with any technical conference paper, your figures and tables should have a clear purpose for existing and be easy to understand. [Include only those figures/tables that provide representative examples that support the point that you want to make in the narrative.](#) Labels, legends, captions, and references from within the body of the text are absolutely necessary.
- **There are no page requirements.** Use whatever space you need to *concisely* provide sufficient details and a solid “story.” A thorough, clear presentation of your ideas will help the reader to effectively assess your work. An unnecessarily long report with meaningless filler material will only irritate the reader.
- Immediately following final project presentations on June 11, I will email feedback to each team on their projects. **Your report should address any concerns regarding the technical content of your work that were identified during your project presentation.** That is, if a deficiency or need for clarification is noted during your project presentation, I will be looking for this information specifically in your project report.
 - Additionally, your presentation is a condensed snapshot of your project report. Thus, all information from your presentation should appear in an expanded form in your project report.

- **Per the course syllabus, project teams must send the following to Prof. Santos via a shared Google Drive or UCLA Box folder by 11:59pm PT on Sat., June 13.** Only a single team representative needs to submit this content.
 - ☐ Folder name having the following form: *X_Report folder*, where *X* is your team #.
 - ☐ Comprehensive, technical project report in PDF format
 - The report must cite any open source code (or prior code from your research lab) that was not written specifically for the course project.
 - You may cite a course project GitHub repo (if it exists) in the report.
 - ☐ Project code in native format
 - ☐ Final project presentation in .ppt format (already downloaded from Google Slides, if necessary)
 - ☐ Supplemental files, such as REQUIRED videos in .mp4 format
 - Optional: You may upload a single representative project video to YouTube and share the YouTube weblink if you would like me to incorporate your projects into a YouTube playlist for a “UCLA MAE C263C Final Project Showcase” that will be open to all.
 - Do not provide YouTube weblinks alone; the original video files enable me to incorporate your projects as exemplars in future MAE C263C course presentations.
- In addition to the Google Drive (or UCLA Box) submission, you will **use the [P.7] “Turnitin” weblink in the “Project resources” folder of the Bruin Learn course website** to upload the following components that have been converted into a searchable PDF format and combined into a single PDF file by 11:59pm PT on Saturday, June 13. Automated plagiarism checking software will be used to check project reports, code, and final presentations. Only a single team representative needs to submit this content.
 - ☐ Comprehensive, technical project report
 - ☐ Project code -- Only custom code written specifically for the course project needs to be printed to a PDF and submitted as part of the Turnitin submission.
 - ☐ Final project presentation
- **Each team member is required to submit a confidential Project Effort Evaluation Form** that rates the contributions of themselves as well as other team members **in .docx or PDF form via Gradescope by 11:59pm PT on Sat., June 13**. This form will be used to adjust project grades on an individual basis if some team members did not carry their weight or went above and beyond the call of duty.
 - *Final project report grades will be lowered by 5% on an individual basis if students do not submit a confidential Project Effort Evaluation Form.*
 - For upload to Gradescope, save your project effort evaluation form with a filename in the following form: *X_Lastname.pdf*, where *X* is your team # and *Lastname* is your last name.
- **Comments on academic integrity** (refer to the course syllabus for the complete version)
 - Soliciting help from online sources including (but not limited to) generative AI tools like ChatGPT, Copilot, and other large language model (LLM) technologies to complete your work in this course is discouraged, but not prohibited. It is unrealistic for me to expect that students and professionals would not use such free tools that are available to them. However, consider that...
 - Contributions from any outside source including (but not limited to) articles, generative AI tools, etc. must be clearly cited and acknowledged in the final submission of any assignment, mini-lab, or report. Points will be deducted for the use of outside sources without proper citation.
 - An important learning outcome of this course is the development of skills to conduct independent research and to communicate technical concepts effectively in oral and/or written form. In the context of the report-based aspects of this course, the use of generative AI tools is similar to receiving outside assistance and raises plagiarism concerns (i.e., that work with your name on it is not actually your own).
 - Project reports and literature reviews are expected to reflect an understanding of key course concepts along with personal effort and independent intellectual thought.
 - LLMs are not immune to “hallucinations” and their responses can be factually incorrect and/or nonsensical. For example, LLMs are not reliable sources when conducting literature reviews. You should verify that sources that you cite actually exist and support your conjectures.